Stream API in Java

=> Stream API :-

➔ Stream API is related to "Collection Framework" or "Group of Objects".

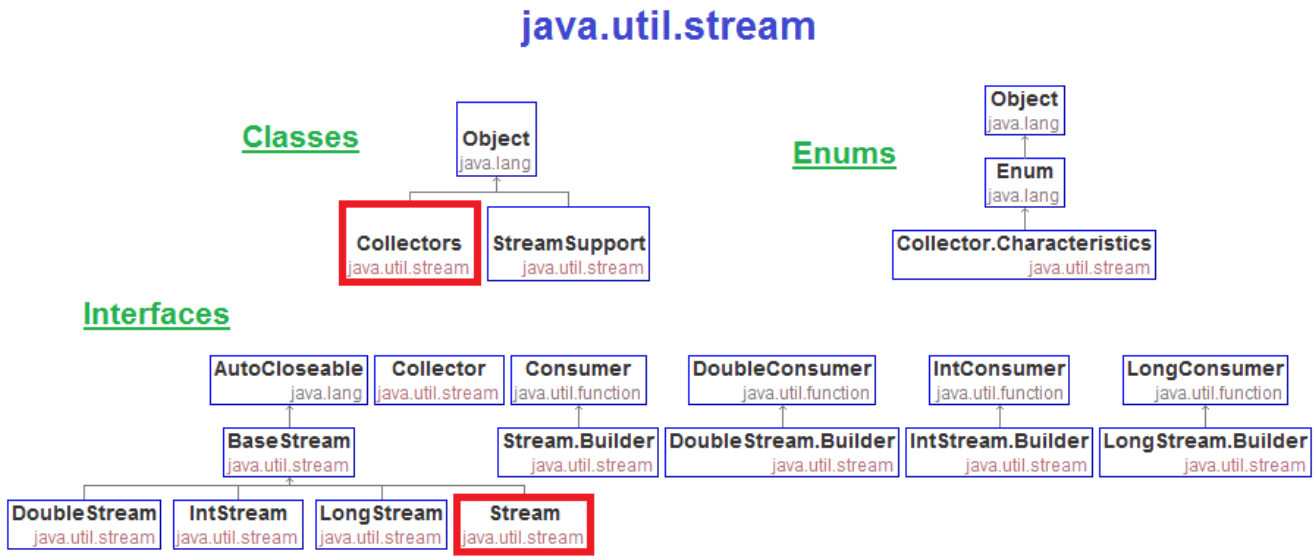
➔ Note : Stream API is not related to Java IO Stream, it is totally different from IO stream.

➔ Stream API was introduced in Java SE 8 version.

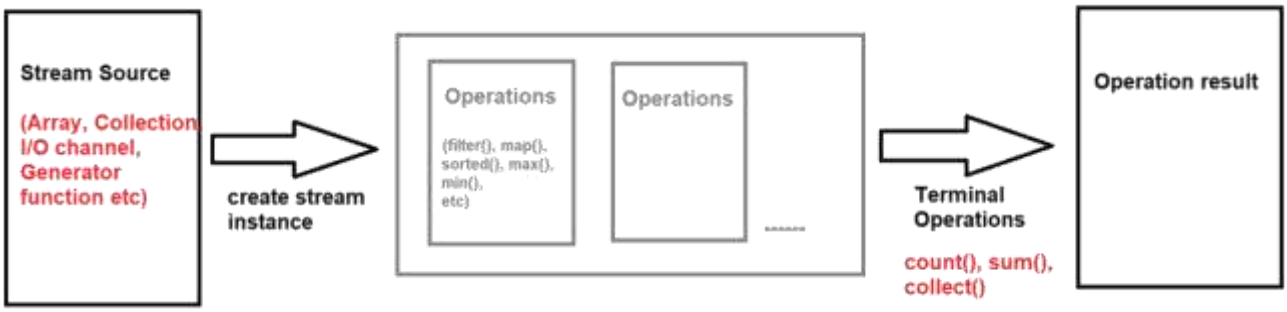
➔ Use of Stream API :-

1. It is used to process or to perform bulk operations on collection objects or group of objects.
2. It decreases the number of lines of code.

➔ Hierarchy of Stream API :-



➔ Stream API Operations Lifecycle :-



1. Get stream from the source and create stream instance.
2. Perform operations(0,1 or many operations) such as filter(), sort(), map() etc and transform it into another stream.
3. Perform terminal operation such as count(), sum() etc and produces the result.

=> Stream Interface :-

-> Stream is an interface which is present in java.util.stream package

-> Syntax :

```
public interface Stream
{
    forEach();
    filter();
    map();
    collect();
    sorted();
    min();
```

```
        max();
        count();

        //some static methods of() {
        - }
        empty() { - }
        builder() { - }
    }
```

=> Task :-

1. WAP to print all the numbers which are greater than 30.
2. WAP to print all the string values whose length is greater than 5.
   a.(deepak, deepesh, amit, rahul, ankit) -> deepak, deepesh
3. WAP to print all the names whose name starts with d.
4. WAP to get the first character of each name.
5. WAP to replace the a charcter with z character.
6. WAP to sort an arraylist containg integer values.
7. WAP to get Minimum and Maximum element in an ArrayList.
8. WAP to print the sum of all the even numbers in an ArrayList